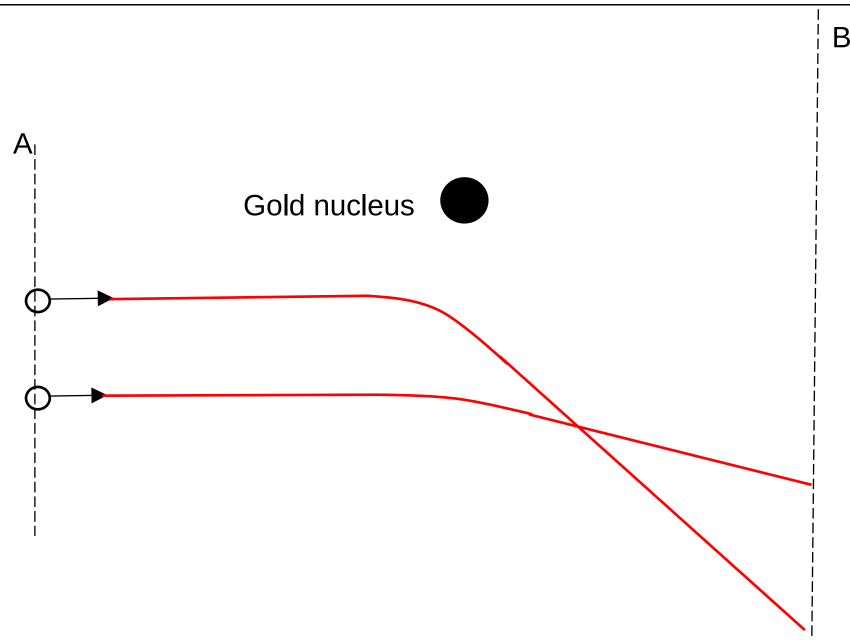
8	**(a)(i)**	To remove air from the chamber, so that alpha particles can successfully travel from the source to the foil and to the screen.	**B1** **B1**
	(a)(ii)	To detect the alpha particles, via scintillations / flashes of light, there must not be other significant sources of light in the room.	**B1** **B1**
	(a)(iii)	Either Observation: A very small fraction of the α-particles was deflected through very large angles Implication: The nucleus has large mass and (positive) charge Or Observation: Most of the α-particles were hardly deflected Implication: The nucleus is very small in size / Most of the atom is empty space	**B1** **B1**
	(a)(iv)	 Deflection is away from nucleus and the upper alpha particle has greater deflection angle Change in direction occurs before the point directly below nucleus Sharp deflection – max of [1]	**B1** **B1**
	(b)(i)	$E = m c^2$ $= (1.0 \times 1.66 \times 10^{-27})\ (3.0 \times 10^{8})^2$ $= 1.494 \times 10^{-10}$ J $= 1.494 \times 10^{-10} / 1.6 \times 10^{-19}$ eV $= 934$ MeV	**M1** **M1** **A0**

	(b)(ii)	Energy absorbed to separate a nucleus into its constituent free/unbound protons and neutrons Or Energy released to form a nucleus from its constituent free/unbound protons and neutrons	**B1** **B1**
	(b)(iii)	Mass defect of Zr = $(40 \times 1.0073) + (57 \times 1.0087) - (97.0980)$ $= 97.7879 - 97.0980$ $= 0.6899$ u B.E. = $0.6899 \times 934 = 644.37$ MeV B.E. per nucleon = 644.37 / 97 = 6.64 MeV/nucleon	**C1** **C1** **A1**
	(c)(i)	Spontaneous decay means that the rate of decay is not affected by external factors such as pressure, temperature, etc. Or Decay occurs on its own, without external stimuli	**B1**
	(c)(ii)	The curve has local fluctuations / jagged edges	**B1**
	(c)(iii)	1. Gamma rays/photons 2. Alpha-particles 3. Gamma 4. Beta-particles **All four correct – [2]** **Subtract [1] for any wrong answer. Minimum is [0]**	
	(c)(iv)	The energy released is shared by the daughter nucleus, beta particle and a neutrino / anti-neutrino.	**B1**